Week 2 Project

DIstro Selections

I kept it simple and used mint and fedora this week. I figured it would be to my advantage to keep using these and getting more comfortable. I ran a check in terminal using “-–version” to confirm they were each using their expected package managers.

```
minty@fresh:~$ apt --version
apt 2.8.3 (amd64) (Mint wrapper)
```

```
zer0cool@fedora:~$ dnf --version
dnf5 version 5.2.17.0
```

App Selection

For app selection, I wanted some things I would use day to day. I like showing off cool things in the terminal, I have found terminal screensavers like pipes and “sl” (ascii train that zooms across terminal) but the coolest I found on alternativeto.net was cmatrix. It makes your terminal look like its from the movie the matrix. I noticed after week one that there was firefox as the default browser on all 4 of the distros I tried so I just wanted anything different and found chromium so I'm going to try that out. Week one I found freetube while poking around on the mint software store. I found it on the website and read a little about it. Youtube is my most used thing on a computer or phone so finding a better version would be cool. Lastly I thought i would need something to record or livestream with so obs got put on the list.

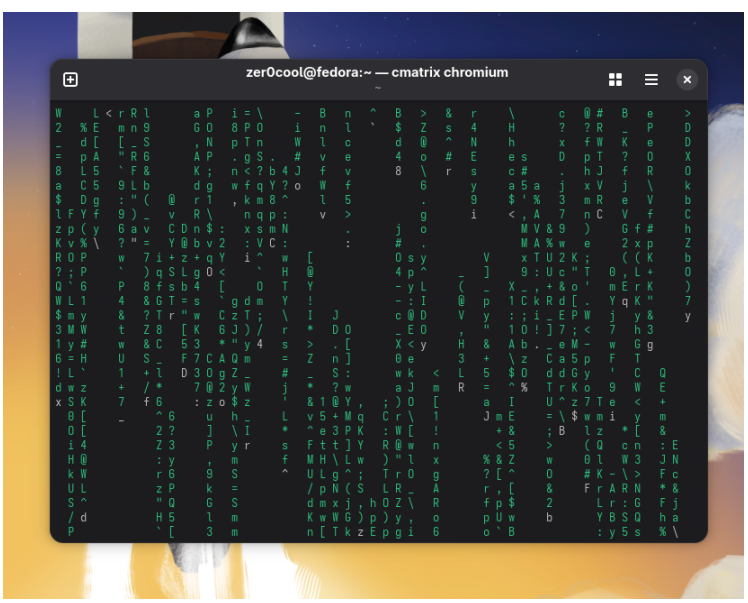
Command Line Installation

(dnf)

I started with fedora and was able to install chromium and cmatrix without an issue

```
zer0cool@fedora:~$ sudo dnf install cmatrix chromium

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for zer0cool:
Updating and loading repositories:
Repositories loaded.
Package                 Arch    Version                 Repository        Size
Installing:
 chromium               x86_64  144.0.7559.109-1.fc43   updates      302.0 MiB
 cmatrix                x86_64  2.0-12.fc43             fedora        76.0 KiB
Installing dependencies:
 chromium-common        x86_64  144.0.7559.109-1.fc43   updates       69.8 MiB
 google-crc32c          x86_64  1.1.2-12.fc43           fedora        28.0 KiB
 libXNVCtrl             x86_64  580.82.09-1.fc43        fedora        41.2 KiB

Transaction Summary:
 Installing:        5 packages

Total size of inbound packages is 125 MiB. Need to download 125 MiB.
After this operation, 372 MiB extra will be used (install 372 MiB, remove 0 B).
Is this ok [y/N]: y
[1/5] cmatrix-0:2.0-12.fc43.x86_64      100% |  97.3 KiB/s |  38.8 KiB |  00m00s
[2/5] google-crc32c-0:1.1.2-12.fc43.x86 100% | 336.2 KiB/s |  22.2 KiB |  00m00s
[3/5] libXNVCtrl-0:580.82.09-1.fc43.x86 100% | 347.1 KiB/s |  23.9 KiB |  00m00s
[4/5] chromium-common-0:144.0.7559.109- 100% |   5.6 MiB/s |  15.8 MiB |  00m03s
[5/5] chromium-0:144.0.7559.109-1.fc43. 100% |  21.6 MiB/s | 109.1 MiB |  00m05s
--------------------------------------------------------------------------------
[5/5] Total                             100% |  21.1 MiB/s | 125.1 MiB |  00m06s
Running transaction
Importing OpenPGP key 0x31645531:
 UserID     : "Fedora (43) <fedora-43-primary@fedoraproject.org>"
 Fingerprint: C6E7F081CF80E13146676E88829B6066316455 31
 From       : file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-43-x86_64
Is this ok [y/N]: y
The key was successfully imported.
[1/7] Verify package files
[2/7] Prepare transaction
[3/7] Installing libXNVCtrl-0:580.82.09-1.fc43.x86_64
[4/7] Installing chromium-common-0:144.0.7559.109-1.fc43.x86_64
[5/7] Installing google-crc32c-0:1.1.2-12.fc43.x86_64
[6/7] Installing chromium-0:144.0.7559.109-1.fc43.x86_64
[7/7] Installing cmatrix-0:2.0-12.fc43.x86_64
Complete!
zer0cool@fedora:~$
```

I used “clear” to remove the clutter from my terminal and ran “cmatrix” and boom....we are in...."You get used to it, I don't even see the code, All I see is blonde, brunette, redhead"



I hit “control + c” to exit. and then typed “sudo dnf remove cmatrix chromium”

I removed cmatrix and chromium without a hitch.

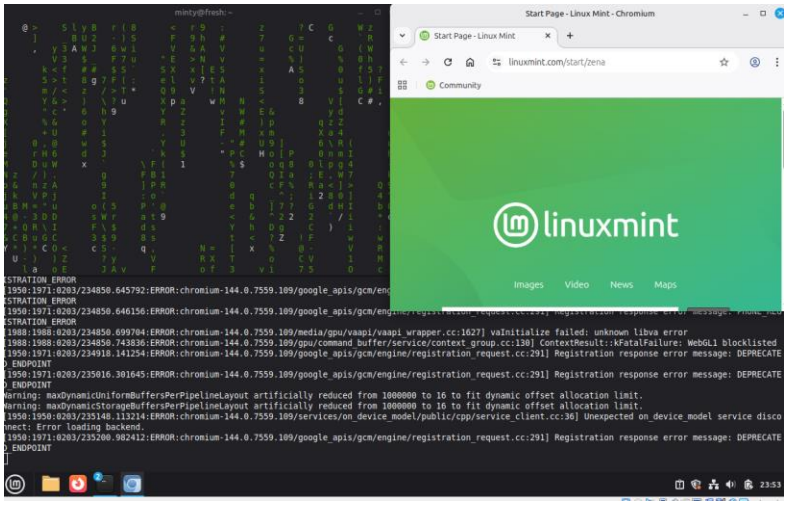
```
Package                 Arch    Version                 Repository        Size
Removing:
 chromium               x86_64  144.0.7559.109-1.fc43   updates      302.0 MiB
 cmatrix                x86_64  2.0-12.fc43             fedora        76.0 KiB
Removing unused dependencies:
 chromium-common        x86_64  144.0.7559.109-1.fc43   updates       69.8 MiB
 google-crc32c          x86_64  1.1.2-12.fc43           fedora        28.0 KiB
 libXNVCtrl             x86_64  580.82.09-1.fc43        fedora        41.2 KiB

Transaction Summary:
 Removing:          5 packages

After this operation, 372 MiB will be freed (install 0 B, remove 372 MiB).
Is this ok [y/N]: y
Running transaction
[1/6] Prepare transaction                100% |   7.0   B/s |   5.0   B |  00m01s
[2/6] Removing chromium-0:144.0.7559.10 100% | 153.0   B/s |  34.0   B |  00m00s
[3/6] Removing chromium-common-0:144.0. 100% |   1.9 KiB/s | 238.0   B |  00m00s
[4/6] Removing libXNVCtrl-0:580.82.09-1 100% |   1.0 KiB/s |   7.0   B |  00m00s
[5/6] Removing google-crc32c-0:1.1.2-12 100% | 321.0   B/s |   9.0   B |  00m00s
[6/6] Removing cmatrix-0:2.0-12.fc43.x86_64                 100% |   3.0   B/s |  19.0   B |  00m05s
Complete!
zer0cool@fedora:~$
```

(apt)

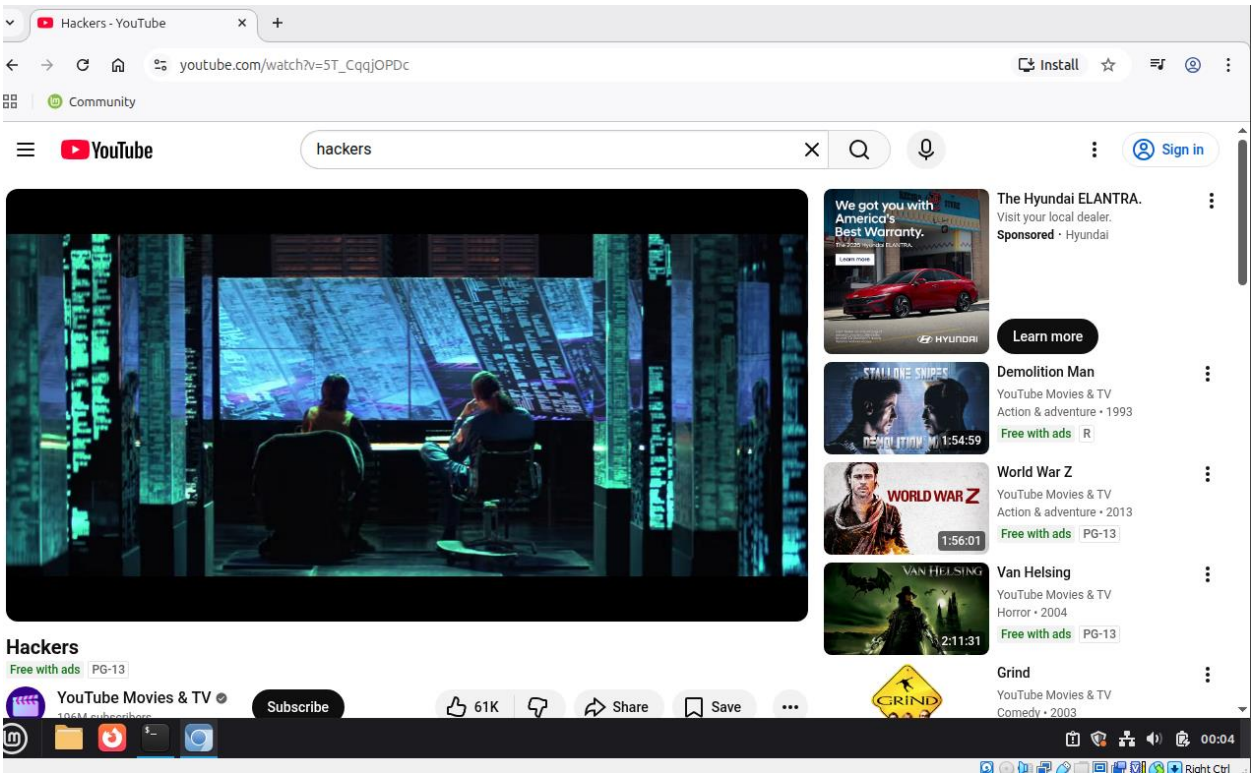
I moved over to mint to install cmatrix and chromium using apt package manager. I ran “sudo apt install chromium cmatrix”

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

minty@fresh:~$ sudo apt install cmatrix chromium
[sudo] password for minty:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  cmatrix-xfont
The following NEW packages will be installed:
  chromium cmatrix
0 upgraded, 2 newly installed, 0 to remove and 101 not upgraded.
Need to get 114 MB of archives.
After this operation, 372 MB of additional disk space will be used.
Get:1 http://packages.linuxmint.com zena/upstream amd64 chromium amd64 144.0.7559.109~linuxmint1+zena [114 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 cmatrix amd64 2.0-6 [33.7 kB]
Fetched 114 MB in 4s (28.9 MB/s)
Selecting previously unselected package chromium.
(Reading database ... 461248 files and directories currently installed.)
Preparing to unpack .../chromium_144.0.7559.109~linuxmint1+zena_amd64.deb ...
Unpacking chromium (144.0.7559.109~linuxmint1+zena) ...
Selecting previously unselected package cmatrix.
Preparing to unpack .../cmatrix_2.0-6_amd64.deb ...
Unpacking cmatrix (2.0-6) ...
Setting up cmatrix (2.0-6) ...
Setting up chromium (144.0.7559.109~linuxmint1+zena) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for mate-menus (1.26.1+mint1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
minty@fresh:~$
```

I typed “chromium” to launch chromium from the terminal. and hit “ctrl + alt + t” to open a new terminal and launched cmatrix from it.



I took chromium for a little test drive and got some hackers playing sense i couldnt get it on freetube.

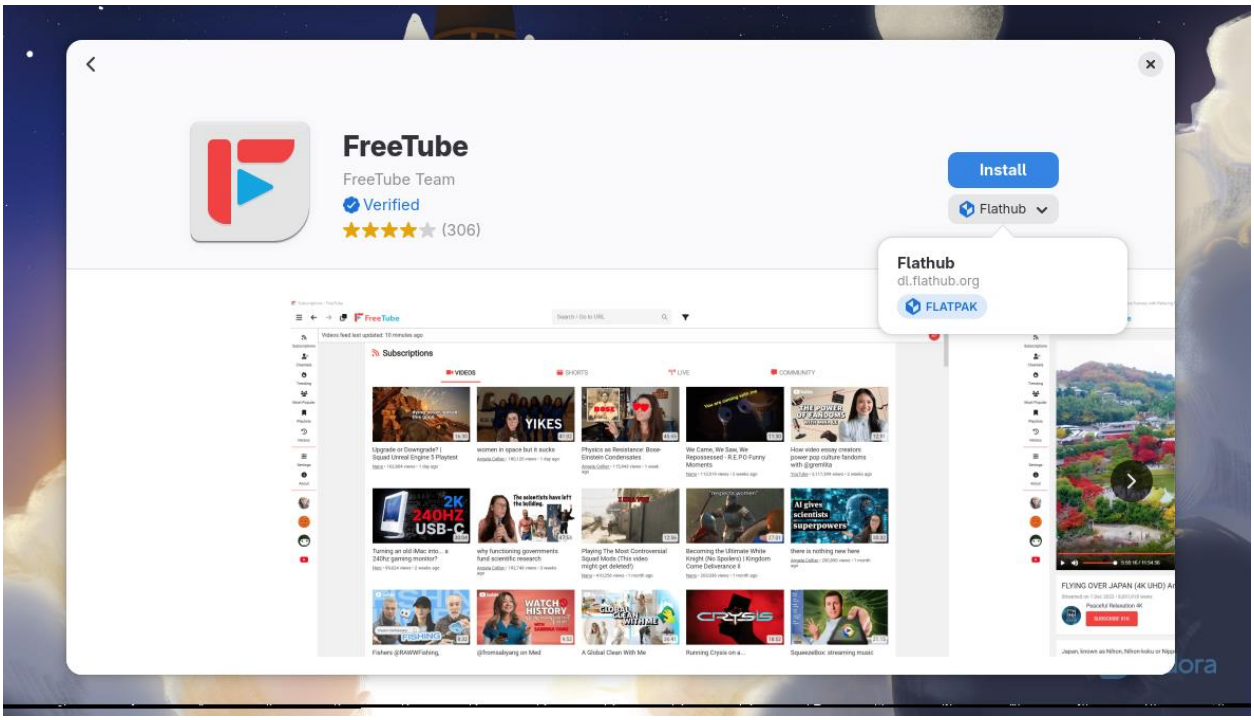


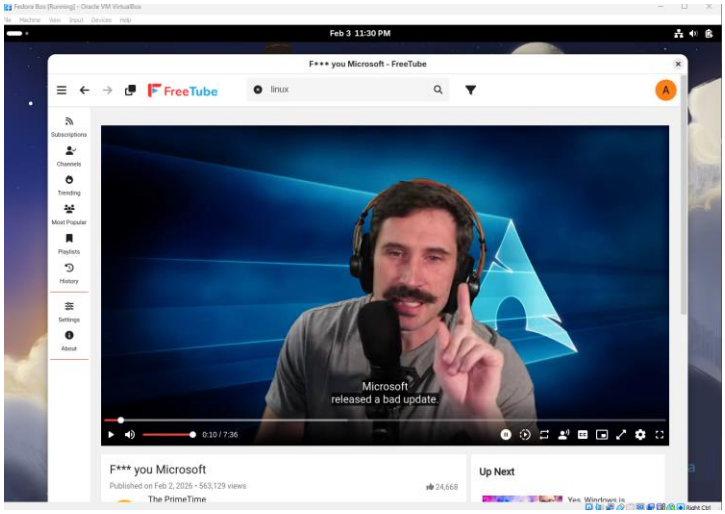
I ran “sudo apt remove cmatrix chromium” and uninstalled cmatrix and chromium.

```
minty@fresh:~$ sudo apt remove cmatrix chromium
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  chromium cmatrix
0 upgraded, 0 newly installed, 2 to remove and 101 not upgraded.
After this operation, 372 MB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 461513 files and directories currently installed.)
Removing chromium (144.0.7559.109~linuxmint1+zena) ...
Removing cmatrix (2.0-6) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for mate-menus (1.26.1+mint1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
minty@fresh:~$ cmatrix
Command 'cmatrix' not found, but can be installed with:
sudo apt install cmatrix
minty@fresh:~$
```

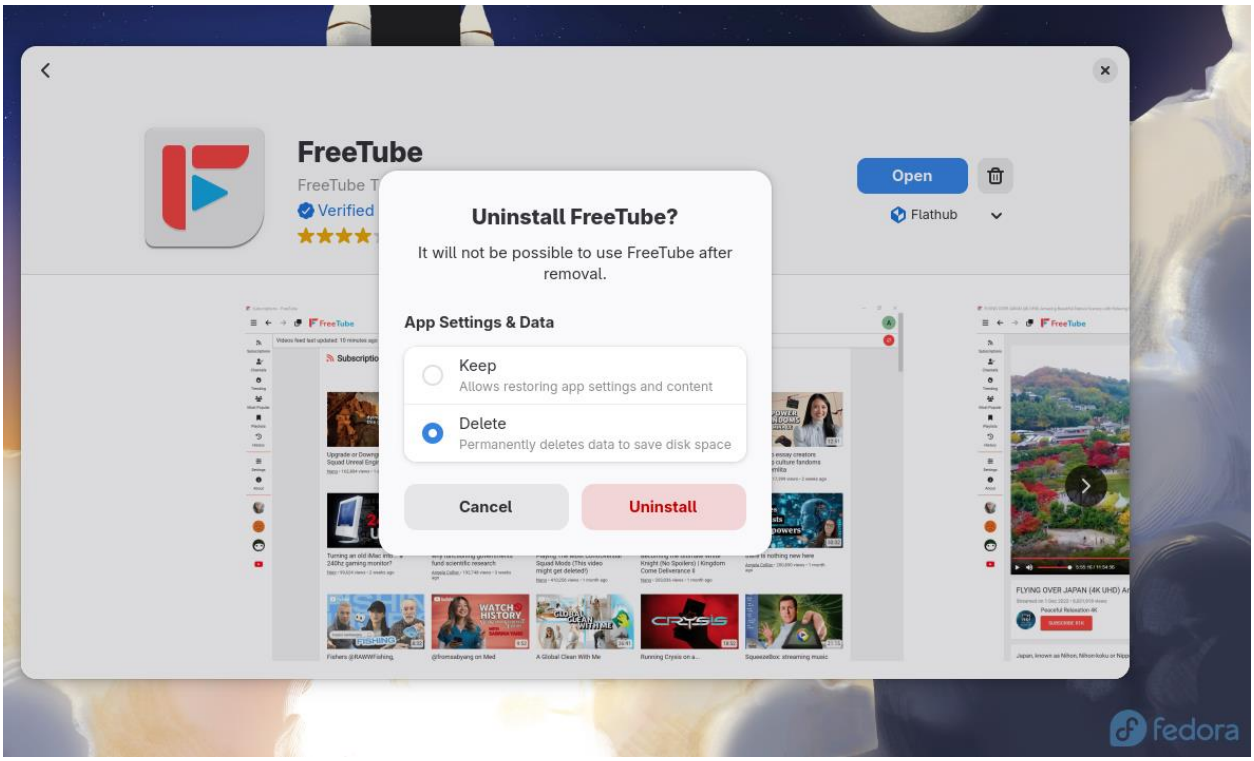
Software Center Installation

In fedora I navigated to the software center and searched for freetube. I found it easily and installed. I wanted to note that it says on the install page that its a flatpak so we could have done this either way. I think this is definitely the easiest option.





Fired up the ole freetube. I will note that I was able to watch most videos with no issue and no ads BUT hackers the movie is on youtube right now in their free movie section and that one wouldn't load because of the system the browser uses to get around the ads. But still not bad for 95% of the stuff I would watch. I went back to the software center and uninstalled it.

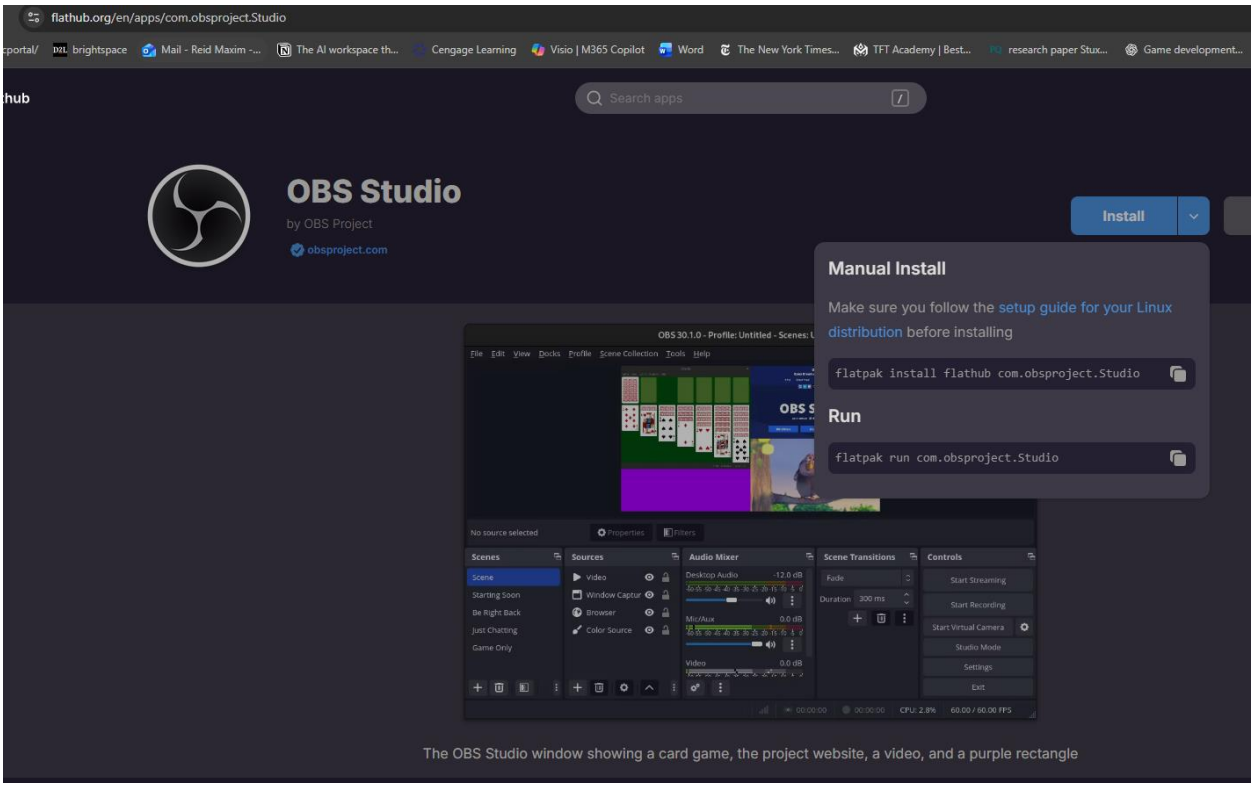


Flatpak Installation

Im starting this portion on mint. I wanted to make sure that I had flatpak installed and ready.

```
minty@fresh:~$ flatpak --version
Flatpak 1.14.6
```

I went to flathub.org and found obs studio listed and they had the install and run code. Right out in the open.



The OBS Studio window showing a card game, the project website, a video, and a purple rectangle

I ran “flatpak install flathub com.obsproject.Studio” and the download started....and kept going ....and going...and goooooooooing. It took forever compared to downloading two apps with apt or dnf.

```
minty@fresh:~$ flatpak --version
Flatpak 1.14.6
minty@fresh:~$ flatpak install flathub com.obsproject.Studio
Looking for matches…
Required runtime for com.obsproject.Studio/x86_64/stable (runtime/org.kde.Platform/x86_64/6.8) found in remote flathub
Do you want to install it? [Y/n]: y

Info: runtime org.kde.Platform branch 6.8 is end-of-life, with reason:
   We strongly recommend moving to the latest stable version of the Platform and SDK
Info: applications using this runtime:
   com.obsproject.Studio

com.obsproject.Studio permissions:
    ipc                  network       fallback-x11       pulseaudio       wayland       x11       devices       file access [1]
    dbus access [2]

    [1] host, xdg-config/kdeglobals:ro, xdg-run/pipewire-0
    [2] com.canonical.AppMenu.Registrar, org.a11y.Bus, org.freedesktop.Flatpak, org.freedesktop.Notifications, org.kde.KGlobalSettings,
        org.kde.StatusNotifierWatcher, org.kde.kconfig.notify, org.kde.kdeconnect


        ID                                      Branch          Op          Remote          Download
 1. [✓] com.obsproject.Studio.Locale            stable          i           flathub           2.1 kB / 47.0 kB
 2. [✓] org.freedesktop.Platform.GL.default     24.08           i           flathub         145.2 MB / 145.9 MB
 3. [✓] org.freedesktop.Platform.GL.default     24.08extra      i           flathub          24.2 MB / 145.9 MB
 4. [✓] org.freedesktop.Platform.openh264       2.5.1           i           flathub         913.7 kB / 971.4 kB
 5. [✓] org.gtk.Gtk3theme.Mint-Y-Aqua           3.22            i           flathub          73.1 kB / 114.9 kB
 6. [✓] org.kde.Platform.Locale                 6.8             i           flathub          18.8 kB / 409.7 MB
 7. [✓] org.kde.Platform                        6.8             i           flathub         322.1 MB / 396.0 MB
 8. [✓] com.obsproject.Studio                   stable          i           flathub         205.6 MB / 212.3 MB

Installing 8/8… ################### 100%  7.6 MB/s  00:00
^[[32;59Rminty@fresh:~$
```

After typing a million different versions of obs i realized its not that simple to open from

terminal. I did open it from the menu on taskbar. But i found this on the flathub website “flatpak run com.obsproject.Studio”

I run “flatpak uninstall com.obsproject.Studio” and uninstalled obs

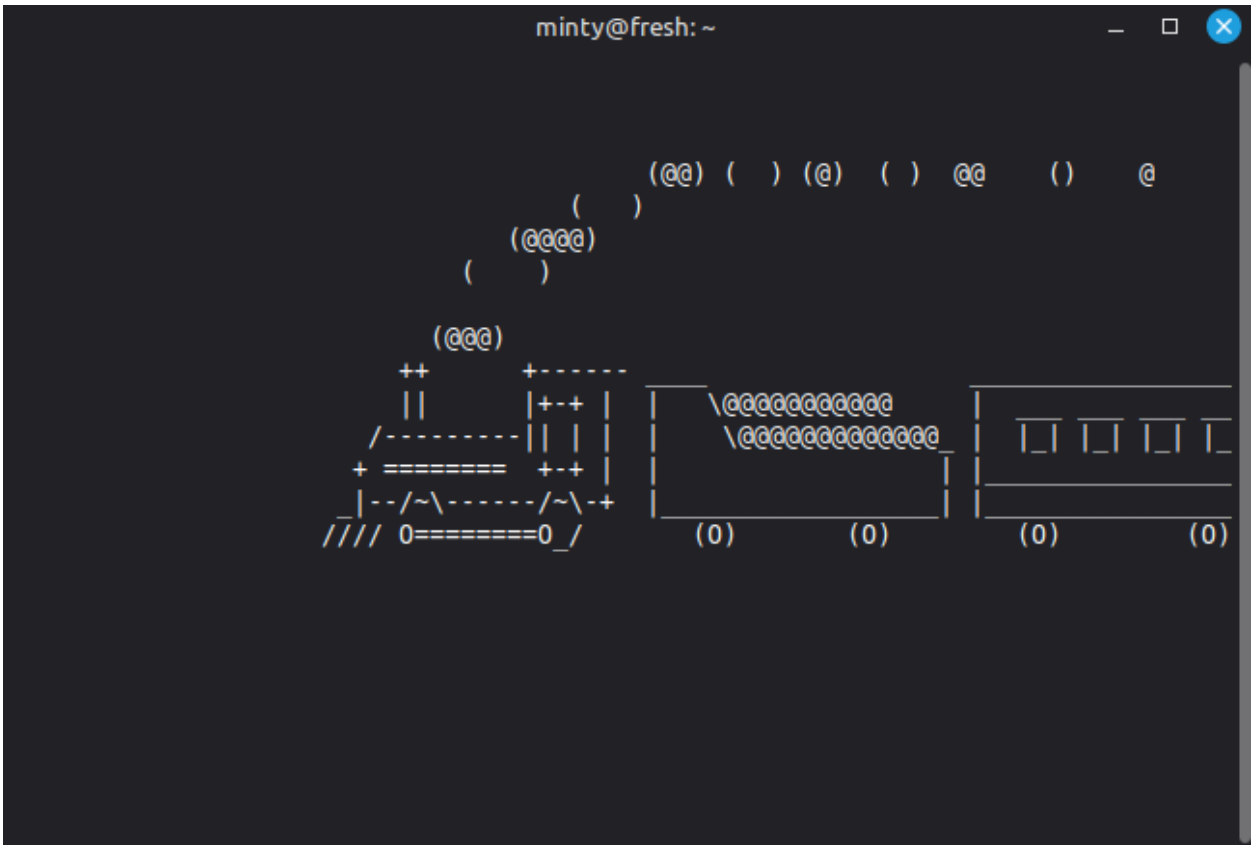
```
minty@fresh:~$ flatpak uninstall com.obsproject.Studio


        ID                                      Branch        Op
 1. [-] com.obsproject.Studio                   stable        r
 2. [-] com.obsproject.Studio.Locale            stable        r
 3. [-] org.kde.Platform.Locale                 6.8           r
 4. [-] org.kde.Platform                        6.8           r

Uninstall complete.
minty@fresh:~$
```

Terminal Commands

I used clear multiple times during the install and unistall process to keep my terminal cleared out between tasks. to start I ran each of them after a “clear”

```
minty@fresh: ~
minty@fresh:~$ echo "week 2 terminal check"
week 2 terminal check
minty@fresh:~$ who
minty    tty7         2026-02-03 23:45 (:0)
minty@fresh:~$ whoami
minty
minty@fresh:~$ id
uid=1000(minty) gid=1000(minty) groups=1000(minty),4(adm),24(cdrom),27(sudo),30(
dip),46(plugdev),100(users),105(lpadmin),125(sambashare)
minty@fresh:~$ date
Wed Feb  4 12:30:01 AM EST 2026
minty@fresh:~$ uname -a
Linux fresh 6.14.0-37-generic #37~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Nov 20
10:25:38 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
minty@fresh:~$ pwd
/home/minty
minty@fresh:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

then gave it a “reset” to start fresh. and then hit the mythical “sl” for a quick ascii train



Reflection

After trying everything out I think the software center was the simplest. I think the average user would not have many issues or complaints using it. Between apt and dnf I didnt notice much difference, dnf showed more on the screen after I put the install command in. I prefer apt out of the two. Most of that reasoning is because it's what I have used the most and I wouldn't be upset if I had to give up apt for dnf. For flatpak I wasn't the biggest fan of the process. Once I googled it and found the website it was pretty straightforward but I don't like how I cant just type a simple straightforward command to download something. I feel like I could learn the layout of the flatpak install command and could have some luck without having to use their website but I don't think it would be as simple as putting the name of the app after “sudo apt install” it also took FOREVER to download so that was kind of boring. I do think that those are small sacrifices to make if it means gaining access to an app in a more secure way if its something sketchy or just in general coming from a cybersecurity standpoint containerizing things seems like a good idea when possible.